

Quadratic cylinder energy and integrability of the Syracuse law

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Abstract

Let μ be the stationary probability measure on $\mathbb{Z}_3$ for the random affine maps $x \mapsto (3x+1)/2^a$, where $\mathbb{P}(a = j) = 2^{-j}$ for $j \geq 1$. With respect to additive Haar probability λ , let ρ_k be the density obtained by spreading each depth- k cylinder mass uniformly over that cylinder. We prove

$$\int_{\mathbb{Z}_3} \rho_k^2 d\lambda \leq 24k + 8k^2 \leq 32k^2 \quad (k \geq 1).$$

The proof combines offset injectivity, an elementary count of distinct integer quotients, and normalized geometric second moments. Together with Tao's fine-scale mixing estimate, the bound gives a single density $\rho = d\mu/d\lambda$ in every $L^p(\lambda)$, $1 \leq p < 2$. A support-overlap argument applied to the stationary equation shows that $\rho \notin L^2(\lambda)$. These statements concern Haar integrability of the random affine law.

1 The stationary law and the results

Write λ for additive Haar probability on $\mathbb{Z}_3$, so $\lambda(x + 3^k\mathbb{Z}_3) = 3^{-k}$. For $a \geq 1$, put

$$\Phi_a(x) = \frac{3x+1}{2^a}.$$

If $a_1, a_2, \dots$ are independent with $\mathbb{P}(a_i = j) = 2^{-j}$, the series

$$X = \sum_{j=0}^{\infty} 3^j 2^{-(a_1 + \dots + a_{j+1})} \tag{1}$$

converges in $\mathbb{Z}_3$. Its law μ is stationary for the maps Φ_a : separating the first exponent in the series gives $X \stackrel{d}{=} \Phi_{a_1}(X')$, where X' has law μ and is independent of a_1 . This is the Syracuse law described in [1, Remark 1.13]. Its cylinder probabilities are

$$p_k(v) = \mu(v + 3^k\mathbb{Z}_3), \quad v \in \mathbb{Z}/3^k\mathbb{Z},$$

and we set

$$\rho_k(x) = 3^k p_k(x \bmod 3^k), \quad N_k = \int \rho_k^2 d\lambda = 3^k \sum_{v \bmod 3^k} p_k(v)^2. \tag{2}$$

In particular, $\rho_0 = 1$ and $N_0 = 1$. All norms and integrals below use the full additive Haar probability λ .

Theorem 1.1. *The Syracuse law has the following properties.*

1. For every integer $k \geq 1$,

$$N_k \leq 24k + 8k^2 \leq 32k^2. \quad (3)$$

2. There is a nonnegative density ρ , with $\mu = \rho\lambda$ and $\int \rho d\lambda = 1$, such that

$$\rho \in L^p(\lambda), \quad \sup_{k \geq 0} \|\rho_k\|_p < \infty \quad \text{for every } 1 \leq p < 2.$$

The same density ρ works for all these exponents.

3. This density does not belong to $L^2(\lambda)$.

The energy proof is independent of the mixing estimate. The density and its subcritical integrability use Tao's Proposition 1.14 [1]. Offset injectivity, which enters the energy argument, is the content of his Lemma 6.2; Section 6 of that paper also uses separation of offsets in collision estimates. We include the elementary decoding and moment arguments needed here.

These estimates are with respect to Haar measure. Embedded ordinary integers form a countable Haar-null set, so the estimates do not supply an integer trace bound or a conclusion about individual Collatz orbits.

2 Words, numerators, and fixed-total fibers

For a word $w = (a_1, \dots, a_k)$ of positive integers, use chronological composition:

$$\Phi_w = \Phi_{a_k} \circ \dots \circ \Phi_{a_1}, \quad A_i = a_1 + \dots + a_i, \quad A_0 = 0, \quad A = A_k.$$

Direct expansion gives

$$\Phi_w(x) = \frac{3^k x + B(w)}{2^A}, \quad B(w) = \sum_{i=0}^{k-1} 3^{k-1-i} 2^{A_i}. \quad (4)$$

Define the positive real translation quantity and the residue

$$Y(w) = \frac{B(w)}{3^k}, \quad r_k(w) = 2^{-A} B(w) \pmod{3^k}. \quad (5)$$

Here the inverse of 2^A is taken modulo 3^k in the second expression. The image $\Phi_w(\mathbb{Z}_3)$ is the cylinder $r_k(w) + 3^k \mathbb{Z}_3$.

The series (1), reduced modulo 3^k , is the endpoint of a length- k composition with the reverse ordering of the exponents. Reversal preserves their product distribution. Consequently,

$$p_k(v) = \sum_{\substack{w \in \mathbb{N}_{>0}^k \\ r_k(w) = v}} 2^{-A(w)}. \quad (6)$$

The sum is absolutely convergent since $\sum_w 2^{-A(w)} = (\sum_{a \geq 1} 2^{-a})^k = 1$.

Lemma 2.1 (Fixed-total decoding). *For fixed $k \geq 1$ and A , the numerator $B(w)$ determines a word w with $A(w) = A$. For a fixed residue v , the map*

$$w \mapsto q(w) := \left\lfloor \frac{B(w)}{3^k} \right\rfloor$$

is therefore injective among words with $A(w) = A$ and $r_k(w) = v$. Moreover, $q(w)$ is a nonnegative integer and $q(w) \leq Y(w)$.

Proof. For $k = 1$, the total fixes the only exponent. For $k \geq 2$, (4) gives

$$B(w) - 3^{k-1} = 2^{a_1} B(a_2, \dots, a_k).$$

The numerator on the right after removing 2^{a_1} is odd: its first summand is odd and its remaining summands are even. Thus $a_1 = v_2(B(w) - 3^{k-1})$. Division recovers the suffix numerator, and induction with total $A - a_1$ recovers the suffix. This is the fixed-total numerator form of offset injectivity [1, Lemma 6.2].

On the fiber with total A and residue v , every numerator has the same remainder modulo 3^k , namely $2^A v$. Equal quotients therefore imply equal numerators and then equal words. The last assertion follows from the definition of the floor. $\square$

Lemma 2.2 (Quotient count). *If $q_1, \dots, q_m$ are distinct nonnegative integers and $Y_j \geq q_j$, then*

$$m^2 \leq 2 \sum_{j=1}^m (1 + Y_j).$$

In particular, if $m_A(v)$ counts any chosen subset of the fixed- (k, A, v) fiber, then

$$m_A(v)^2 \leq 2 \sum_{w \text{ in the chosen fiber}} (1 + Y(w)). \quad (7)$$

Proof. Order the distinct integers increasingly. The j -th is at least $j - 1$, so

$$\sum_{j=1}^m (1 + Y_j) \geq \sum_{j=1}^m j = \frac{m(m+1)}{2} \geq \frac{m^2}{2}.$$

Apply Lemma 2.1 to obtain (7). $\square$

3 The energy estimate

Fix $k \geq 1$. Introduce the probability distribution on words

$$\pi_k(w) = 3^k 4^{-A(w)} = \prod_{j=1}^k (3 \cdot 4^{-a_j}). \quad (8)$$

It is normalized because $\sum_{a \geq 1} 3 \cdot 4^{-a} = 1$. This distribution will be used only to calculate a moment; the cylinder probabilities in (6) retain their fair weights $2^{-A(w)}$.

Lemma 3.1 (Energy reduction). *One has*

$$N_k \leq \frac{4}{k} \mathbb{E}_{\pi_k} [A^2 (1 + Y)]. \quad (9)$$

Proof. First restrict to words with $A \leq M$, where $M \geq k$. For each residue v , denote the corresponding partial sum in (6) by $p_{k,M}(v)$. Every word has $A \geq k$. The elementary estimate

$$\frac{1}{A^2} \leq 2 \left(\frac{1}{A} - \frac{1}{A+1} \right) \quad (A \geq 1)$$

implies $\sum_{A=k}^M A^{-2} \leq 2/k$. Weighted Cauchy–Schwarz and (7) now give

$$\begin{aligned} p_{k,M}(v)^2 &= \left(\sum_{A=k}^M 2^{-A} m_A(v) \right)^2 \\ &\leq \frac{2}{k} \sum_{A=k}^M A^2 4^{-A} m_A(v)^2 \\ &\leq \frac{4}{k} \sum_{\substack{w: A(w) \leq M \\ r_k(w)=v}} A(w)^2 4^{-A(w)} (1 + Y(w)). \end{aligned}$$

Sum over v and multiply by 3^k . Each word has exactly one residue, and hence

$$3^k \sum_v p_{k,M}(v)^2 \leq \frac{4}{k} \sum_{A(w) \leq M} \pi_k(w) A(w)^2 (1 + Y(w)).$$

Let $M \rightarrow \infty$. The left side tends to N_k , and the nonnegative sums on the right increase to the displayed expectation. The next lemma shows that this expectation is finite. $\square$

Lemma 3.2 (Normalized second moments). *For $k \geq 1$,*

$$\mathbb{E}_{\pi_k} [A^2(1 + Y)] \leq 6k^2 + 2k^3. \quad (10)$$

Proof. The identity $\sum_{a \geq 1} a^2 z^a = z(1+z)/(1-z)^3$, for $0 < z < 1$, gives

$$\sum_{a \geq 1} a^2 2^{-a} = 6, \quad \sum_{a \geq 1} a^2 (3 \cdot 4^{-a}) = \frac{20}{9} \leq 6. \quad (11)$$

For $0 \leq i \leq k$, put $C_i = 3^i 2^{-A_i}$. Formula (4) yields

$$Y = \frac{1}{3} \sum_{i=0}^{k-1} C_i^{-1}. \quad (12)$$

Multiplying π_k by C_i^{-1} changes precisely the first i coordinate weights:

$$(3 \cdot 4^{-a}) \frac{2^a}{3} = 2^{-a}.$$

Therefore $\pi_k(w) C_i(w)^{-1}$ is a normalized product distribution, with fair coordinates before position $i+1$ and the weights $3 \cdot 4^{-a}$ thereafter. No normalization factor is lost in this change of measure.

The pointwise inequality $A^2 \leq k \sum_{j=1}^k a_j^2$, together with (11), gives

$$\mathbb{E}_{\pi_k} [A^2 C_i^{-1}] \leq 6k^2 \quad (0 \leq i \leq k).$$

For $i = 0$, this bounds $\mathbb{E}_{\pi_k} A^2$. Summing (12) gives

$$\mathbb{E}_{\pi_k} [A^2 Y] = \frac{1}{3} \sum_{i=0}^{k-1} \mathbb{E}_{\pi_k} [A^2 C_i^{-1}] \leq 2k^3.$$

All the summands are nonnegative, so Tonelli's theorem justifies the sums, and the bounds also establish their finiteness. $\square$

Combining Lemmas 3.1 and 3.2 proves

$$N_k \leq \frac{4}{k} (6k^2 + 2k^3) = 24k + 8k^2 \leq 32k^2,$$

which is part (1) of Theorem 1.1.

4 Mixing and subcritical integrability

The published input is the following consequence of [1, Proposition 1.14, equations (1.26)–(1.27)]: for each $A > 0$ there is a finite constant C_A such that

$$\|\rho_n - \rho_m\|_1 \leq C_A m^{-A} \quad (n \geq m \geq 1). \quad (13)$$

To check the normalization, consistency of the cylinder masses gives

$$\|\rho_n - \rho_m\|_1 = \sum_{v \bmod 3^n} |p_n(v) - 3^{m-n} p_m(v \bmod 3^m)|.$$

This is exactly the oscillation in the cited proposition.

It follows that ρ_k is Cauchy in $L^1(\lambda)$. Let ρ be a nonnegative representative of its limit. The limit has integral one. For every cylinder D of depth m , consistency gives $\int_D \rho_k d\lambda = \mu(D)$ when $k \geq m$. Passing to the L^1 limit proves the same identity for ρ . Cylinders generate the Borel sigma algebra, so the finite measures $\rho\lambda$ and μ are equal.

Fix $1 < p < 2$, and put $\delta_k = \rho_{k+1} - \rho_k$ for $k \geq 1$. By (13),

$$\|\delta_k\|_1 \leq C_A k^{-A}. \quad (14)$$

The elementary inequality $(u - v)^2 \leq 2(u^2 + v^2)$ and the energy bound give

$$\int \delta_k^2 d\lambda \leq 2(N_{k+1} + N_k) \leq 64((k+1)^2 + k^2) \leq 320k^2. \quad (15)$$

Hölder interpolation between the first and second moments yields

$$\begin{aligned} \|\delta_k\|_p &\leq \|\delta_k\|_1^{(2-p)/p} \left(\int \delta_k^2 d\lambda \right)^{(p-1)/p} \\ &\leq C_A^{(2-p)/p} 320^{(p-1)/p} k^{[-A(2-p)+2(p-1)]/p}. \end{aligned} \quad (16)$$

For completeness, the first inequality follows by writing $|u|^p = |u|^{2-p}(|u|^2)^{p-1}$, applying Hölder with conjugate exponents $1/(2-p)$ and $1/(p-1)$, and taking the p -th root.

Choose a positive integer

$$A > \frac{4p-2}{2-p}.$$

Then the exponent in (16) is at most -2 . With $D_p = C_A^{(2-p)/p} 320^{(p-1)/p}$, finite telescoping and Minkowski give, for every $n \geq 1$,

$$\|\rho_n\|_p \leq \|\rho_1\|_p + \sum_{k=1}^{n-1} \|\delta_k\|_p \leq \|\rho_1\|_p + 2D_p. \quad (17)$$

Here ρ_1 is a bounded cylinder function, and $\sum_{k \geq 1} k^{-2} \leq 2$ follows from the same telescoping estimate used in the energy proof.

The already established L^1 convergence implies convergence in measure to ρ . The Fatou property for convergence in measure therefore gives

$$\|\rho\|_p \leq \liminf_{n \rightarrow \infty} \|\rho_n\|_p \leq \|\rho_1\|_p + 2D_p.$$

Indeed, one may choose a subsequence realizing the $\liminf$, pass to a further subsequence converging almost everywhere, and apply Fatou's lemma to the p -th powers. This step uses the same L^1 limit for each p . The case $p = 1$ was established above, and $\rho_0 = 1$ supplies the remaining finite level. This proves part (2) of Theorem 1.1.

5 The stationary equation and the critical exponent

Let

$$J(y) = \frac{3y+1}{2}, \quad D = J(\mathbb{Z}_3) = \{x \in \mathbb{Z}_3 : x \equiv 2 \pmod{3}\}.$$

Multiplication by 2 preserves λ , while $J_*\lambda = 3\lambda|_D$. For a measurable function h , write its extension along this branch as

$$(Lh)(x) = \begin{cases} h((2x-1)/3), & x \in D, \\ 0, & x \notin D. \end{cases}$$

Thus, whenever the integral is defined,

$$\int Lh \, d\lambda = \frac{1}{3} \int h \, d\lambda, \quad (18)$$

and, for integrable h, g ,

$$\|Lh - Lg\|_1 = \frac{1}{3} \|h - g\|_1. \quad (19)$$

The geometric weights yield the exact finite-level recursion

$$\rho_{k+1}(x) = \frac{1}{2} \rho_{k+1}(2x) + \frac{3}{2} (L\rho_k)(x) \quad (k \geq 0). \quad (20)$$

To see this, separate $a = 1$ from $a \geq 2$ in the outermost exponent of the random series. The first event has probability $1/2$ and applies J to the depth- k tail. Conditional on $a \geq 2$, the exponent $a - 1$ has the original geometric law, and the output is half the corresponding depth- $(k+1)$ variable. The two changes of variables have Haar density factors 3 and 1, respectively, which gives (20).

Passing to the L^1 limit in (20), using the invariance under doubling and (19), proves

$$\rho(x) = \frac{1}{2} \rho(2x) + \frac{3}{2} (L\rho)(x) \quad \text{for almost every } x. \quad (21)$$

For an explicit estimate, the L^1 norm of the difference between the two sides is at most

$$\frac{3}{2} \|\rho_{k+1} - \rho\|_1 + \frac{1}{2} \|\rho_k - \rho\|_1,$$

which tends to zero.

We use the following finite-measure observation.

Lemma 5.1 (Positive support overlap). *Let f, a, b be nonnegative measurable functions on a finite measure space (Ω, ν) . Suppose $f = a + b$ almost everywhere,*

$$\nu\{a > 0\} = \nu\{f > 0\}, \quad 0 < \int b \, d\nu < \infty,$$

and ab is integrable. Then $\int ab \, d\nu > 0$.

Proof. Since $a \leq f$ almost everywhere, $\{a > 0\}$ is contained in $\{f > 0\}$ up to a null set. Equality of their finite measures makes these sets equal up to a null set. If $\int ab \, d\nu = 0$, nonnegativity and integrability imply $ab = 0$ almost everywhere. But wherever $b > 0$, the equation $f = a + b$ gives $f > 0$, hence $a > 0$ outside a null set. This forces $b = 0$ almost everywhere, contradicting its positive integral. $\square$

Suppose now that $\rho \in L^2(\lambda)$, and put

$$a(x) = \frac{1}{2}\rho(2x), \quad b(x) = \frac{3}{2}(L\rho)(x).$$

Both functions belong to L^2 , their product is integrable, and (21) gives $\rho = a + b$ almost everywhere. Doubling preserves Haar measure, so

$$\lambda\{a > 0\} = \lambda\{\rho > 0\}.$$

Equation (18) and $\int \rho d\lambda = 1$ give $\int b d\lambda = 1/2$. Lemma 5.1 therefore implies $\int ab d\lambda > 0$.

On the other hand, the same changes of variables yield

$$\int a^2 d\lambda = \frac{1}{4} \int \rho^2 d\lambda, \quad \int b^2 d\lambda = \frac{3}{4} \int \rho^2 d\lambda.$$

Squaring $\rho = a + b$ and integrating would then give

$$\int \rho^2 d\lambda = \int \rho^2 d\lambda + 2 \int ab d\lambda,$$

a contradiction. This proves part (3) of Theorem 1.1. Since λ is a probability measure, $L^p(\lambda) \subseteq L^2(\lambda)$ for every real $p \geq 2$. Thus, among real exponents $p \geq 1$, the density belongs to $L^p(\lambda)$ exactly when $p < 2$. The support argument uses no ergodicity statement and no claim that ρ is positive almost everywhere on a prescribed ambient set.

Formalization record

The following table records the corresponding Lean endpoints. The energy proof uses the native reverse ordering of the word exponents; an explicit reversal identity connects its numerator to (4). Its moment proof uses the same geometric second moments through a direct recursion on native words.

Statement	Lean endpoint	Recorded status
Native quadratic energy	<code>FairEnergy.residue_energy_le_linear_quadratic</code>	Strict compile and kernel check passed
Haar quadratic energy	<code>canonicalPadicRho_sq_integral_le_thirtytwo_quadratic</code>	Strict compile and kernel check passed
One subcritical density	<code>canonicalSyracuseMeasure_has_subcritical_density</code>	Strict compile and kernel check passed
Uniform finite-level L^p , $1 < p < 2$, $n \geq 1$	<code>exists_canonicalPadicRho_uniform_eLpNorm_bound</code>	Strict compile and kernel check passed
Actual stationary equation	<code>canonicalHaarDensity_stationary</code>	Strict compile and kernel check passed
Failure of L^2	<code>canonicalHaarDensity_not_memLp_two</code>	Strict compile and kernel check passed
One density with the exact L^p range	<code>canonicalSyracuseMeasure_has_sharp_lp_density</code>	Strict compile and kernel check passed

Here `FairEnergy` abbreviates `CollatzCylinderPacking.Arithmetic.FairEnergy`; the other displayed endpoints are in `Erdos1135.Tao`. The uniform-bound endpoint is public; the extension to $n = 0$ and $p = 1$ uses the finite norm at depth zero and the mass-one identity at every depth. The accompanying bundle freshly compiled all 362 internal source modules and checked each with the official Lean kernel checker. Its complete declaration audit was also compiled and kernel checked, for 726 successful commands. The audit covers 10,465 module/constant rows, 10,445 distinct names and 736 encoded private rows; only `propext`, `Classical.choice` and `Quot.sound` occur as axioms. Complete external input inventories agreed before and after the run. The external Lean and Mathlib libraries were fingerprinted and reused; their complete source closures were not rebuilt. Subsequent setup reuse and record verification in fresh processes, including a relocated directory, also passed. The bundle preserves the earlier failed strict-compilation attempt and the 32 reviewed compatibility patches. The fresh dependency-download branch was not exercised.

Use of AI

OpenAI Codex was used extensively for mathematical exploration, development of the Lean proofs and verification scripts, automated review, and preparation and revision of this manuscript. The accompanying formal artifacts have the verification scope recorded above. Automated review does not constitute external peer review or imply review or endorsement by the authors of the cited works.

References

- [1] T. Tao, Almost all orbits of the Collatz map attain almost bounded values, *Forum of Mathematics, Pi* **10** (2022), e12. doi:10.1017/fmp.2022.8. Author manuscript: arXiv:1909.03562v7.